

Max Fierro

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ACADEMIC HISTORY

University of California, Berkeley

Aug. 2021 – Dec 2025

B.A. Computer Science. GPA: 3.4

Berkeley, CA

Selected Coursework: *Operating Systems, Deep Neural Networks, Machine Learning, Abstract Linear Algebra, Probability Theory, Signal Processing, Advanced Algorithms, Database Systems, Computer Architecture.*

PROFESSIONAL EXPERIENCE

Software Engineer Intern (Experimentation Systems)

May 2025 – Aug. 2025

Meta Platforms, Inc.

New York, NY

- Migrated **status reporting** to modular experimentation infrastructure (serving products like [ServiceLab](#)).
- Constructed **distributed workflows** under SLA of eventual consistency and retry-based fault tolerance.
- Oversaw **safe migration rollout** by designing and monitoring system health signals during deployment.
- Authored **15-page RFC proposal** of state log pipeline and model-predictive controller for workflows.

Software Engineer Intern (ML Infrastructure)

May 2024 – Aug. 2024

Meta Platforms, Inc.

New York, NY

- Developed **systems for ML** in Production Model Training team, improving online training efficiency.
- **Eliminated redundant parameter copying** in generic post-training pipeline, reducing infrastructure cost of online model productionization (which is how ads ranking models are served, for example).
- Integrated **uninformed black-box optimization** for large hyper- and production parameter spaces.
- Introduced **adaptive experiment-based optimization** in collaboration with research scientists.

Software Engineer Intern (Ads Bidding Products)

May 2023 – Aug. 2023

Meta Platforms, Inc.

Menlo Park, CA

- Implemented pre-alpha **advertiser value optimization** feature, allowing the assignment of contrastive weights to **customizable user subpopulations** in ad campaign objectives (used during ad ranking).
- Contributed to the feature's **data schema and system design** via comprehensive feasibility studies.
- Expanded **Meta Ads Manager** web interface to publish new bidding data to global ad delivery system.

MetaU Engineering Intern (Early Career)

June 2022 – Aug. 2022

Meta Platforms, Inc.

Menlo Park, CA

- Learned **large-scale application design** with the WhatsApp Data Archive and Transport team.
- Participated in a bootcamp-style course on Obj-C and Swift **iOS development** using UIKit and SwiftUI.
- Built zero-to-one **task management application** supporting a DAG of task-subtask relationships.

Lead Endpoint Engineer

May 2022 – Feb. 2024

UC Berkeley Student Affairs IT

Berkeley, CA

- **Led** an endpoint management team responsible for the security and accessibility of more than 400 devices.
- Provided advising to **UC Berkeley IT leaders and CIO** as part of the Student Technology Council.
- Maintained campus partnerships and **oversaw organizational work** including the hiring of 4 engineers.

SELECTED PROJECTS

[GamesmanNova](#) | Abstract sequential game analysis Rust system.

- Parallel system for **efficient exploration of graphs** induced by game rulesets, facilitating dataset generation.
- Implements runtime-monomorphic support for **generic N-player games** (including puzzles as 1-player games).
- Multi-database design for **access pattern specialization** for algorithmic workloads under varying game sizes.

Technical Articles | Long-form works on topics I am passionate about.

- Derivation of the Elo system via parameter likelihood gradient ascent on gaussian-distributed performance.
- Techniques for large-scale state-based dynamic programming, emphasizing game representation and solving.

TECHNICAL SKILLS

General: Technical leadership, mathematical maturity, principled design, technical writing and communication.

Technologies: UNIX systems, relational DBMSs, RPC frameworks, PyTorch, NumPy, Git, Mercurial, $\text{\LaTeX}$.

GPPLs: Rust (most experience), Python, C/C++, Java, Swift, JavaScript (least experience).

Updated September 10, 2025